PROJECT TOPIC -: VEHICLE PARKING MANAGEMENT

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1. Introduction / Problem Statement

I will be working on a **web application** which will be managing the valet parking details of vehicles entering a parking lot. This web application will manage vehicle entry, exit, and parking assignments, providing real-time management for valet operations. The entire application will fully be deployed on **AWS cloud infrastructure**.

2. Project Objectives

- Frontend will be created using CSS, HTML and bootstrap.
- Databases (Amazon RDS or Dynamo DB) will be integrated to store vehicle entry, exit and payment data.
- CI/CD will be implemented using AWS **CodePipeline**, **CodeBuild** and **CodeDeploy**.
- For security will be using **AWS WAF** (Web Application Firewall), **IAM role** for authorisation and **cloudwatch** for monitoring.

Workflow Summary:

1. Admin enters vehicle details via the website.
2. Data is stored in Amazon RDS.(vehicle details, check in and check out)
3. On check-out the final bill will be calculated on hourly basis and slip will be generated.
4. Aws CodePipeline automatically updates the new version of code push from GitHub to S3 and redeploys the backend.
5. CloudWatch and WAF will monitor traffic and will provide the logs.

3. Tools & Technologies

CI/CD: AWS Codepipeline, Codebuild, Codedeploy
Frontend Technologies: HTML, CSS, JavaScript, Bootstrap
Backend Technologies : AWS EC2 (Node.js or Python)
Database Resource:AWS RDS (MySQL/PostgreSQL)
Security & Monitoring AWS WAF, IAM, Cloudwatch
Version Control: GitHub